Imperfections in Solids

In this topic we will try to answer the following questions:

- What types of defects arise in solids?
- Are these defects undesirable?
- How do defects affect material properties?
- Can the number and type of defects be varied and controlled?

Imperfections in Solids

There is no such thing as a perfect crystal!

- Thermodynamically "impossible"
- "defects" lower the energy of a crystal & make it more stable
- always have vacancies and impurities, to some extent

Defect does not necessarily imply a bad thing, examples:

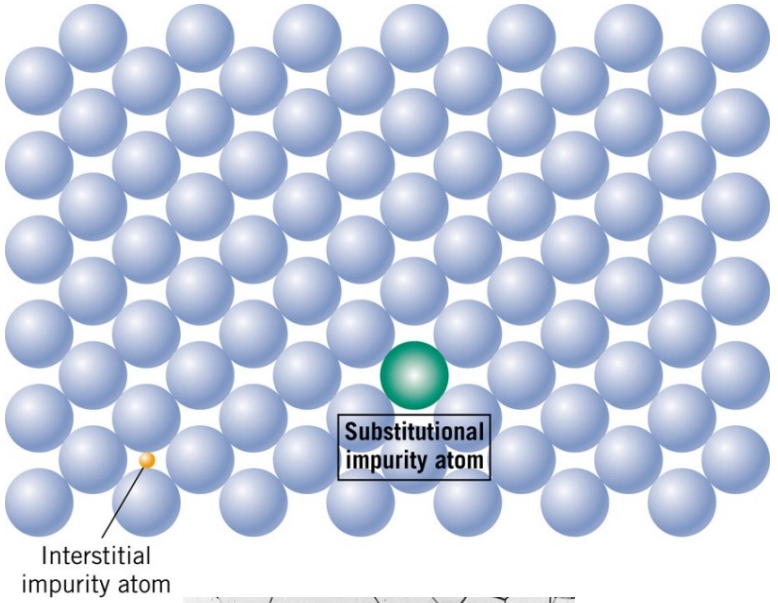
- addition of C to Fe to make steel
- addition of Cu to Ni to make thermocouple wires
- addition of Ge to Si to make thermoelectric materials
- addition of Cr to Fe for corrosion resistance
- introduction of *grain boundaries* to strengthen materials

***"Defect"** (in this context) can be either desirable or undesirable. In general, a defect simply refers to a disruption in the crystalline order of an otherwise periodic material.*

Types of Imperfections

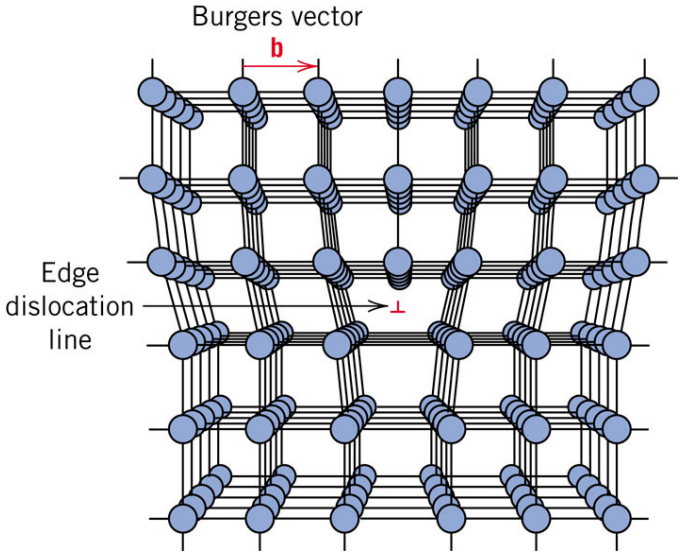
1- Point defects:

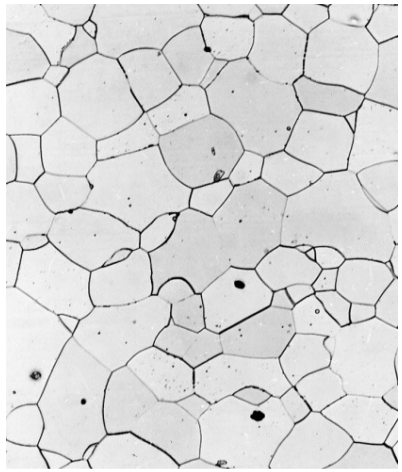
- *Vacancy atoms*
- *Interstitial atoms*
- *Substitutional atoms*



2- Line defects

- *Dislocations*

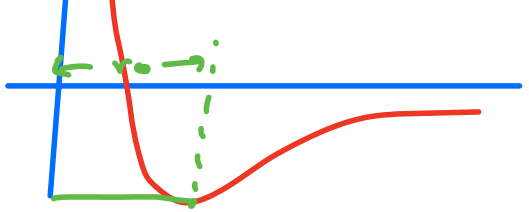




3- Area defects:

- *Grain Boundaries*

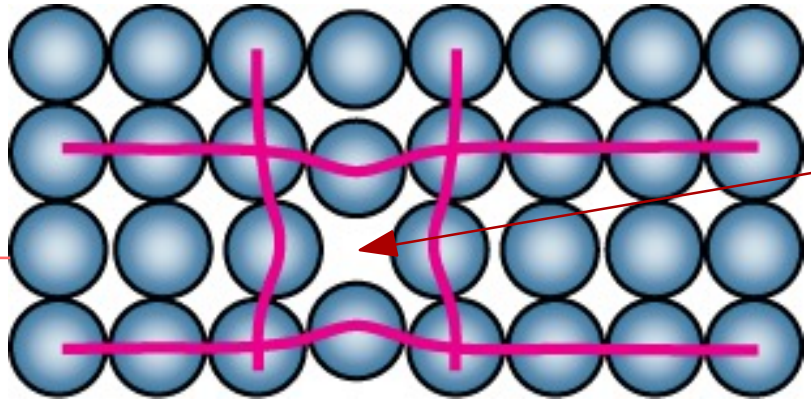
Point Defects



vacancy → a lattice site that is missing an atom:

All crystals contain some vacancies.

distortion of planes



missing atom

Vacancy

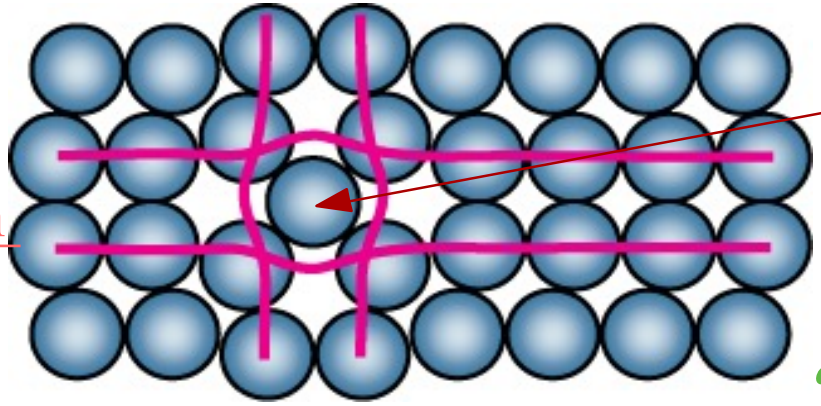
missing atom

self-interstitial → an atom from the crystal that crowds its way into an otherwise empty void between atoms

distorts the planes and almost destroys the order

↑E req.

distortion of planes



self-interstitial

atom squeezed inside

*Self-interstitials are far less common than vacancies because of the relatively large **energy** required to squeeze an atom into the small voids between existing sites.*

Equilibrium Concentration: *Point Defects*

- Equilibrium concentration varies with temperature!

$$\frac{N_D}{N} = \exp\left[\frac{-Q_D}{kT}\right]$$

No. of defects

No. of potential defect sites.

Activation energy

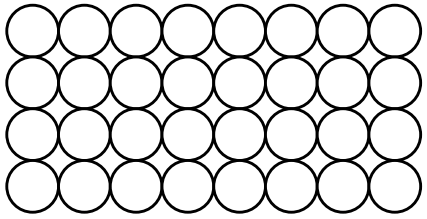
E_{req} for the activation of the defect

Temperature

Boltzmann's constant

(1.38×10^{-23} J/atom K)

(8.62×10^{-5} eV/atom K)



Each lattice site is a potential vacancy site

This is called an *Arrhenius* equation, after Svante August Arrhenius, a Swedish chemist who won the *Nobel* prize in chemistry in 1903.

Observing Equilibrium Vacancy Concentration

- Increasing temp. causes surface island of atoms to grow.
- Why? ↑T = ↑Area

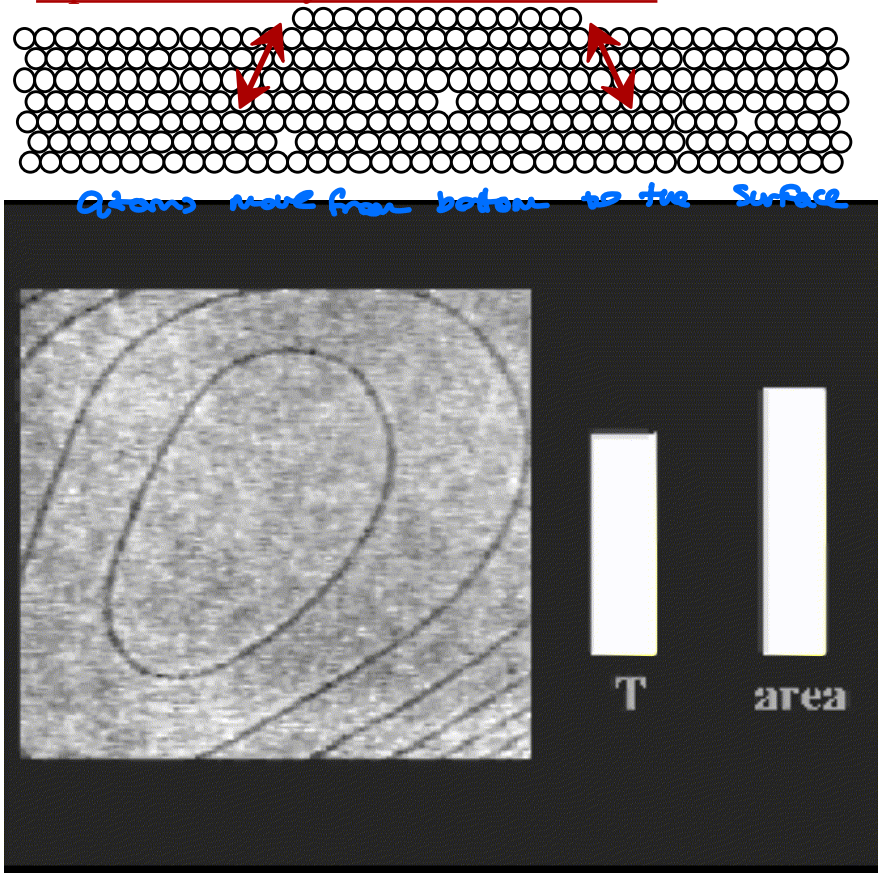
- *Because the equil. vacancy concentration increases via atom motion from the crystal to the surface, where they join the island.*

So, can we observe this?

Yes!

For e.g. the low energy electron microscope view of a (110) surface of NiAl shows how the surface islands grow with temp.

Island grows/shrinks to maintain equil. vancancy conc. in the bulk.

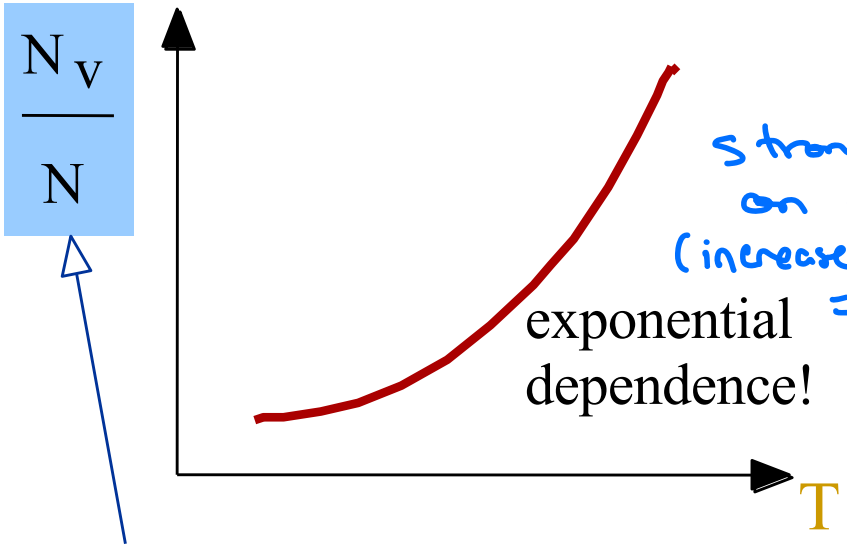


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Measuring Activation Energy

$$N_v = N_o \exp\left(-\frac{Q}{kT}\right)$$

- We can get Q from an experiment.

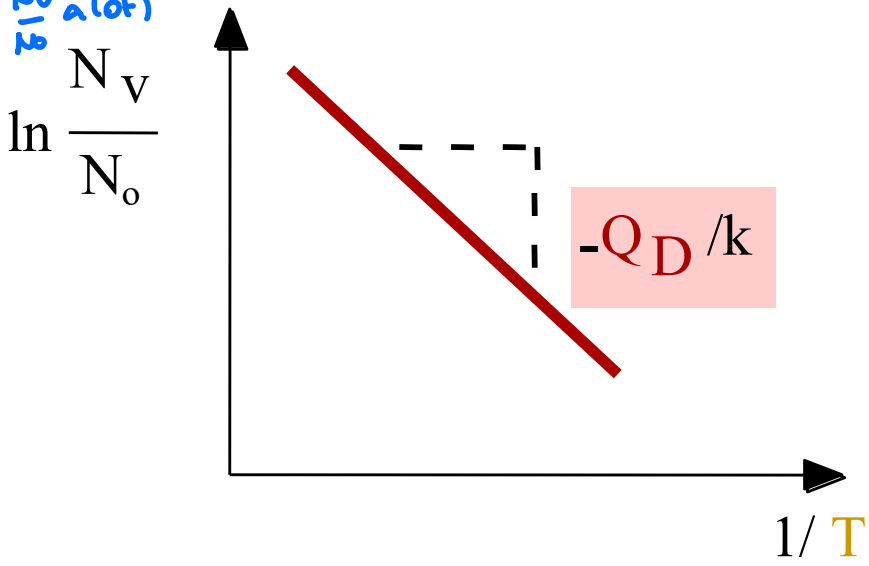


defect concentration

taking logarithms of both sides of the above equation:

$$\ln\left[\frac{N_v}{N_o}\right] = -\frac{Q}{k}\frac{1}{T}$$

plot $\ln\left[\frac{N_v}{N_o}\right]$ versus $\frac{1}{T}$:



Note: exp(-Q/kT) *is a ……. function of temperature*

Example:

Calculate the fraction of atom sites that are vacant for lead at its melting temperature of 327°C (600 K). Assume an activation energy of 0.55 eV/atom.

*To compute the fraction of atom sites that are vacant in lead at 600 K, we must employ **Arrhenius** Equation (eqn. 4.1).*

As stated in the problem, Qv = 0.55 eV/atom. Thus,

$$\frac{N_v}{N_0} = \exp\left(-\frac{Q_v}{kT}\right) = \exp\left[-\frac{0.55\ \text{eV/atom}}{(8.62\times10^{-5}\ \text{eV/atom}\cdot\text{K})(600\text{K})}\right] = 2.41\times10^{-5}$$

Point Defects

impurity → addition of an atom of a different species than the "host" or matrix

- Alloys – other types of atoms are deliberately added to give the material certain properties
  - *May or may not result in the same crystal structure*
  - *May or may not result in secondary phases*

Example 1: add 1 at.% Sn to Pb

the tin can replace the lead if is a little is added so it keeps the same crystal structure

i.e., of every 100 Pb lattice sites, 1 is occupied by an Sn atom

Result: ..single..phase..alloy

*same **crystal structure** as pure Pb*

Example 2: add 25 at.% Sn to Pb

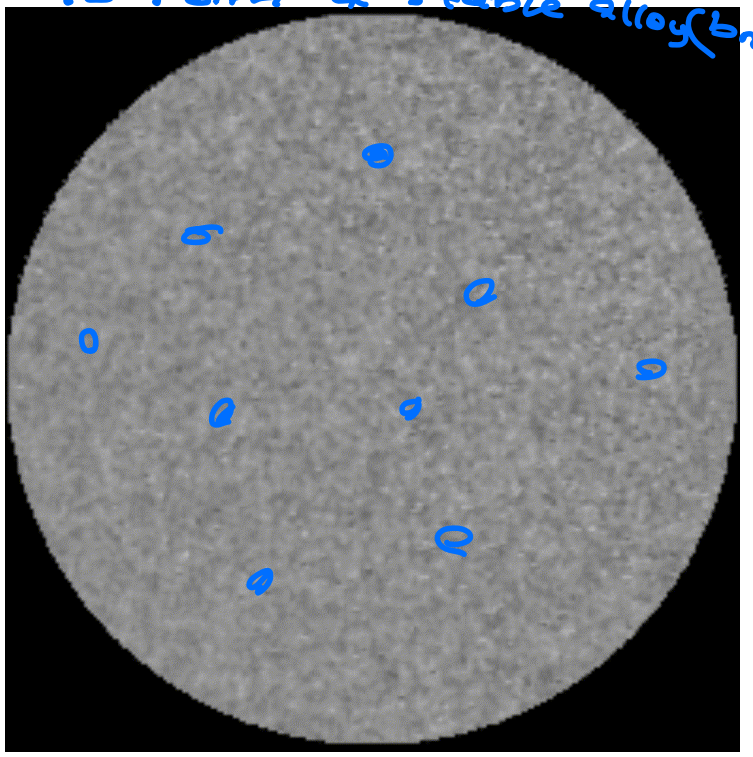
if too much is added it enters second-phase and changes

Result: an alloy with ...two-..phase. microstructure (distinct regions of Sn)

*"**solubility**" of Sn (in the solid state) is exceeded*

Alloying a Surface

- Low energy electron microscope view of a (111) surface of Cu.
- Sn islands move along the surface and alloy the Cu with Sn atoms, to make bronze.
- The islands continually move into unalloyed regions and leave tiny bronze particles in their wake.
- Eventually, the islands disappear.



Reprinted with permission from: A.K. Schmid, N.C. Bartelt, and R.Q. Hwang, "Alloying at Surfaces by the Migration of Reactive Two-Dimensional Islands", Science, Vol. 290, No. 5496, pp. 1561-64 (2000). Field of view is 1.5 μm and the temperature is 290K.

Solid Solution

- A homogeneous distribution of two or more elements.
- "solute" atoms are added without altering the crystal structure or resulting in formation of a new phase.
- Solid solution is a particular type of alloy
- Two types: **substitutional** and **interstitial**

replaces some ↙ Cu in Sn C in Fe ↘ squeezes into pre existing space

"solvent" – the host material, usually the element or compound present in the greatest amount.

"solute" – the minor phase, added to the solvent. Usually the element or compound present in the lower concentration.

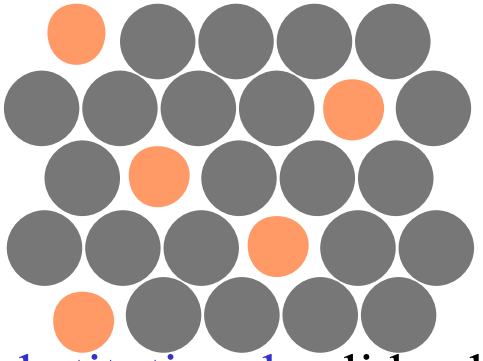
"phase" is a region of uniform composition or crystal structure

What would a solid solution look like?

Point Defects in Alloys

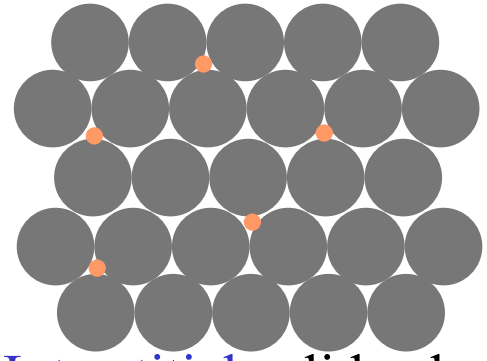
Two outcomes if impurity (B) added to host (A):

- Solid solution of B in A (i.e., random dist. of point defects)



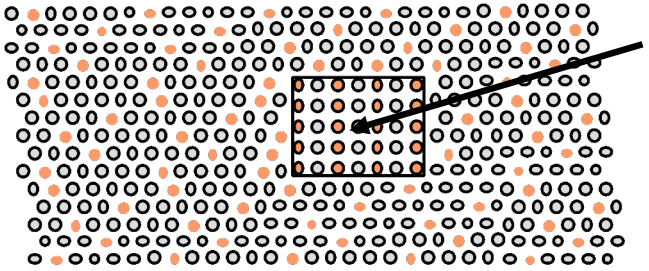
Substitutional solid soln.
(e.g., Cu in Ni)

OR



Interstitial solid soln.
(e.g., C in Fe)

- Solid solution of B in A plus particles of a new phase (usually for a larger amount of B)



Second phase particle
- *different* composition
- *often different* ..structure...

Solid Solution

Not all elements form solid solutions. There are very specific rules that govern the extent to which solid solutions can form.

For ***appreciable*** **substitutional** solubility to occur the following factors must hold:

- Atomic size factor

  the difference in atomic radii between the two atoms must be $\leq \pm 15\%$ (within 15%.)

- Electronegativity

  the difference in electronegativity between the two atoms must be $\leq \pm 0.4$ *eV* (9x)

  (i.e. large differences $\rightarrow$ *compound formation (intermetallics))*

- Crystal structure

  The crystal structure of each element must be the same

- Valence

  For a given solvent, a solute with a higher valency is more likely to be soluble than one of lower valency, all other factors being equal.

Composition and Conversion wt% ↔ at.%

weight % atomic %.

Mixtures or alloys of two (or more) elements, are quantitatively described in terms of composition:

Consider a mixture of B in A:

$$\text{weight percent B} = \frac{\text{mass of B}}{\text{total mass of A + B}} \times 100\% = C_B$$

$$\text{atomic percent B} = \frac{\text{\# moles of B}}{\text{total \# moles of A + B}} \times 100\% = C'_B$$

- Conversion between wt % and at% in an A-B alloy:

$$C_B = \frac{C'_B A_B}{C'_A A_A + C'_B A_B} \times 100 \qquad C'_B = \frac{C_B/A_B}{C_A/A_A + C_B/A_B}$$

- Basis for conversion:

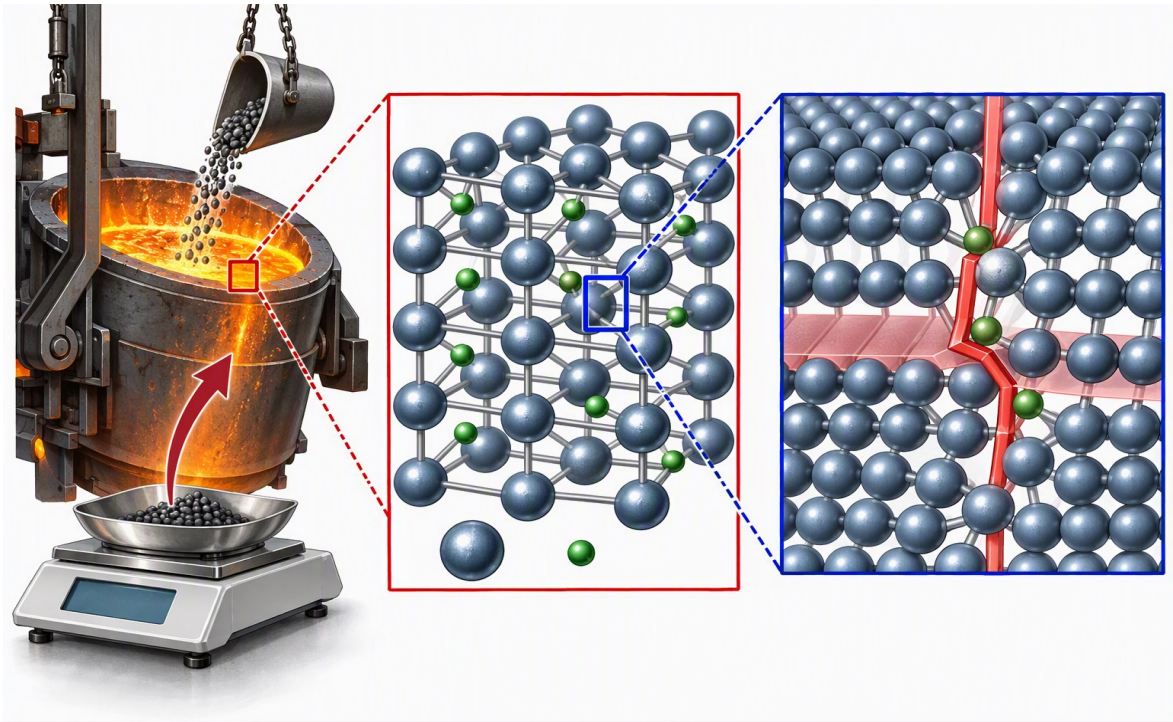
mass of B = moles of B x A_B

mass of A = moles of A x A_A

- *In industry usually use weight % for alloys: easier to add (x) kgs of Mg to (Y) tonnes of Al.*
- *For chemical/scientific measures often need **atomic %***

Why convert wt% to at.%?

- We've understood where solute atoms go.
- We now know how to convert a production recipe into that atomic percentage.



Example
0.80 wt% C steel

Take a 100 g basis:

C: 0.80 g ÷ 12.01 = 0.0666 mol
Fe: 99.20 g ÷ 55.85 = 1.776 mol

at.% C = 0.0666/(0.0666+1.776) × 100

≈ 3.6 at.% C

Meaning: about 3.6 of every 100 atoms are C, mostly in interstitial sites.

- For PROCESSING → weigh alloy additions (wt%)
- To predict properties → often convert to at.% → identify defects / phases (mostly R&D)

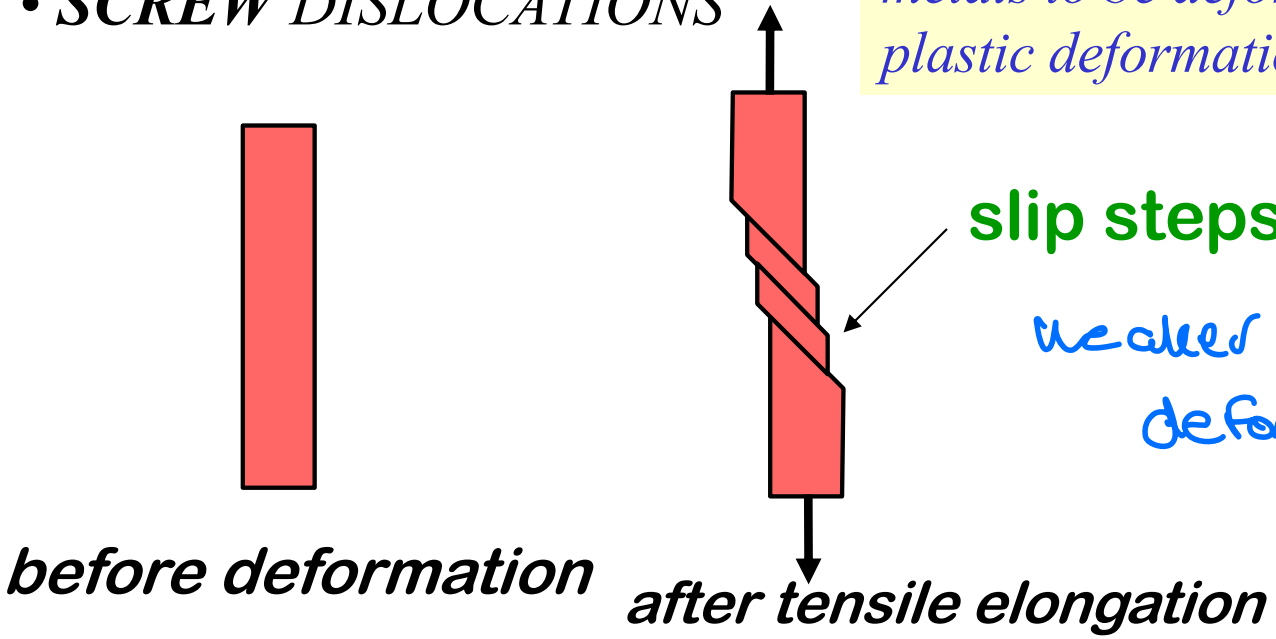
Example: interstitial C distorts the Fe lattice and impedes **dislocation** motion, linking composition to strengthening.

Linear Defects

Dislocations

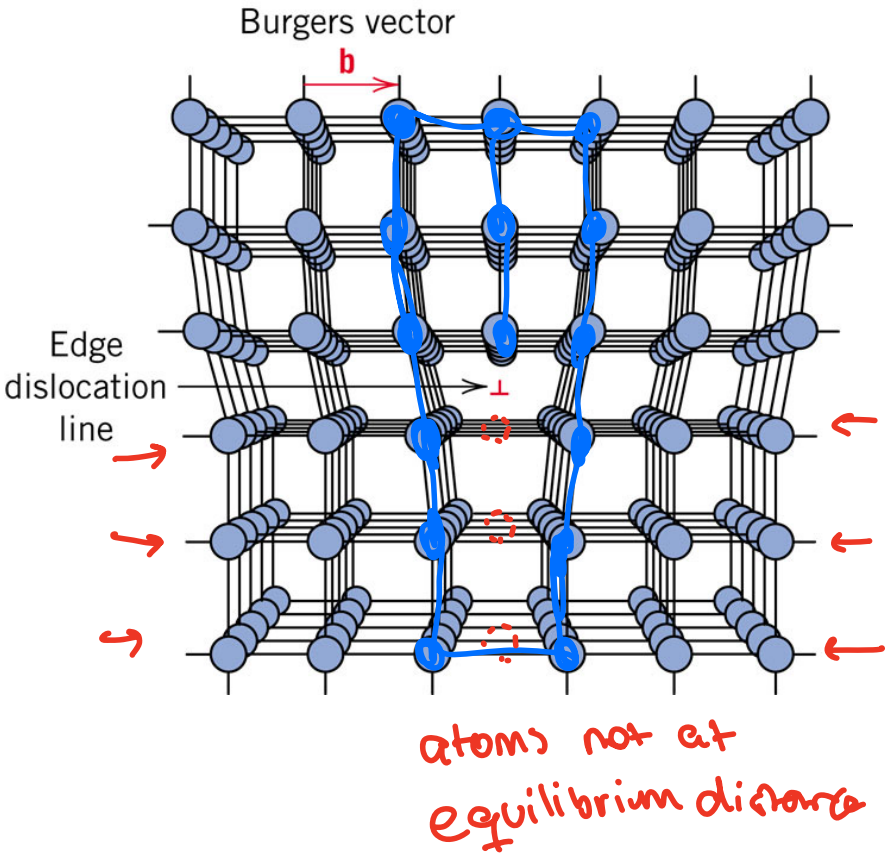
- Dislocations result from solidification from the melt, from mechanical work (*e.g., rolling, drawing, compressive impact, tensile or shear stress*), or from thermal stresses
- It is very difficult to prepare a **dislocation-free** crystal!!!
- 2 Types:
  - ***EDGE*** *DISLOCATIONS*
  - ***SCREW*** *DISLOCATIONS*

Dislocations *make metals weaker than they should be, BUT also allow metals to be deformed (ie. allow plastic deformation). (Chp. 6)*



Edge Dislocation

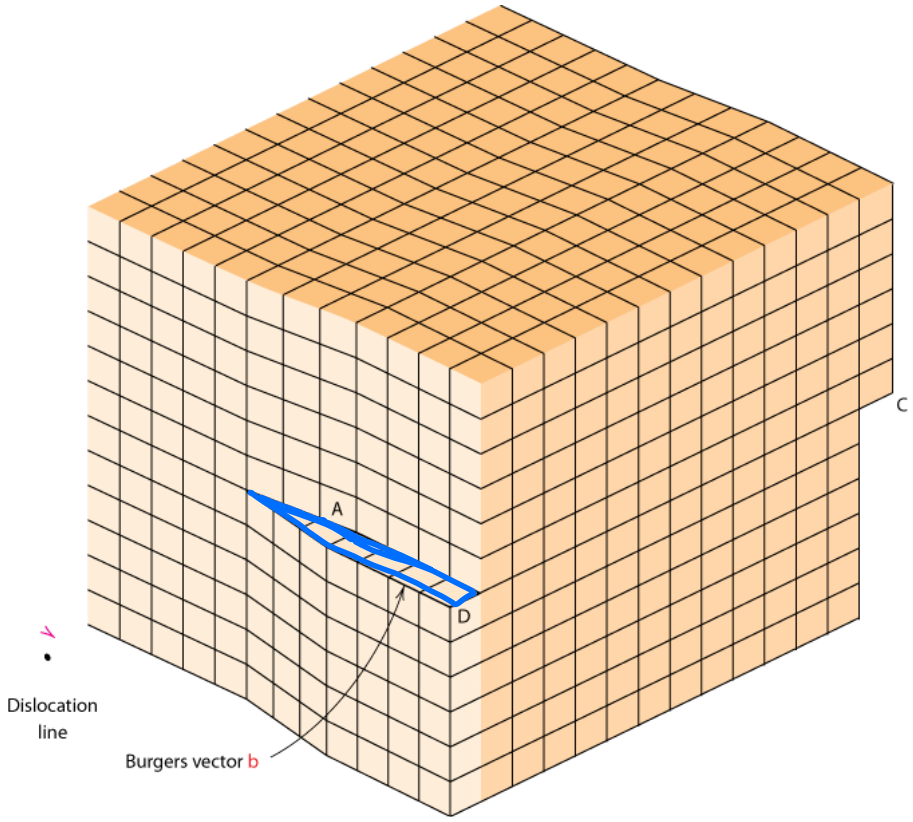
- *Think of edge dislocation as an extra half-plane of atoms inserted in a crystal.*
- Misalignment of atomic planes due to the extra half plane.

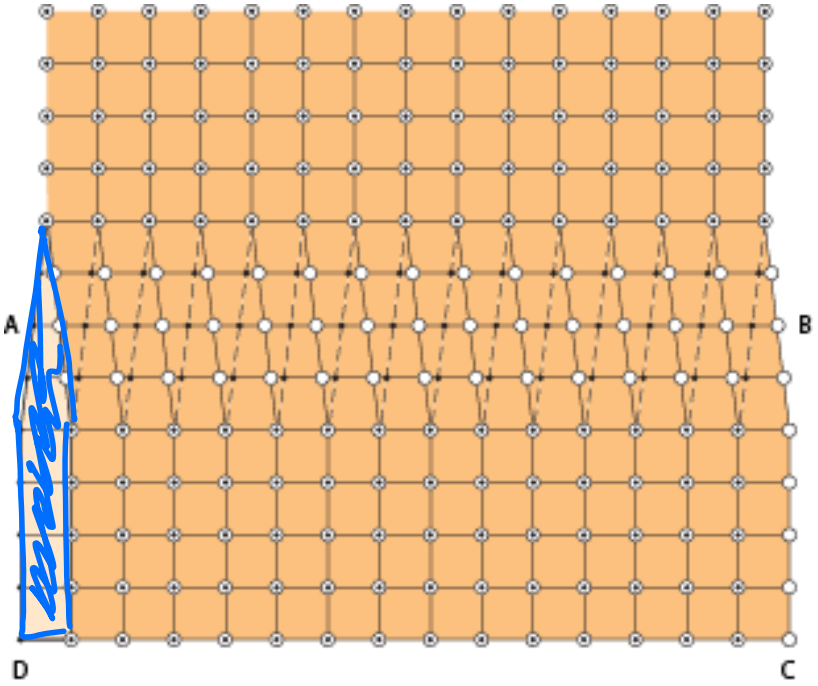


Burger's vector (b) = magnitude + direction of lattice distortion.

Screw Dislocation

can't exist w/out edge

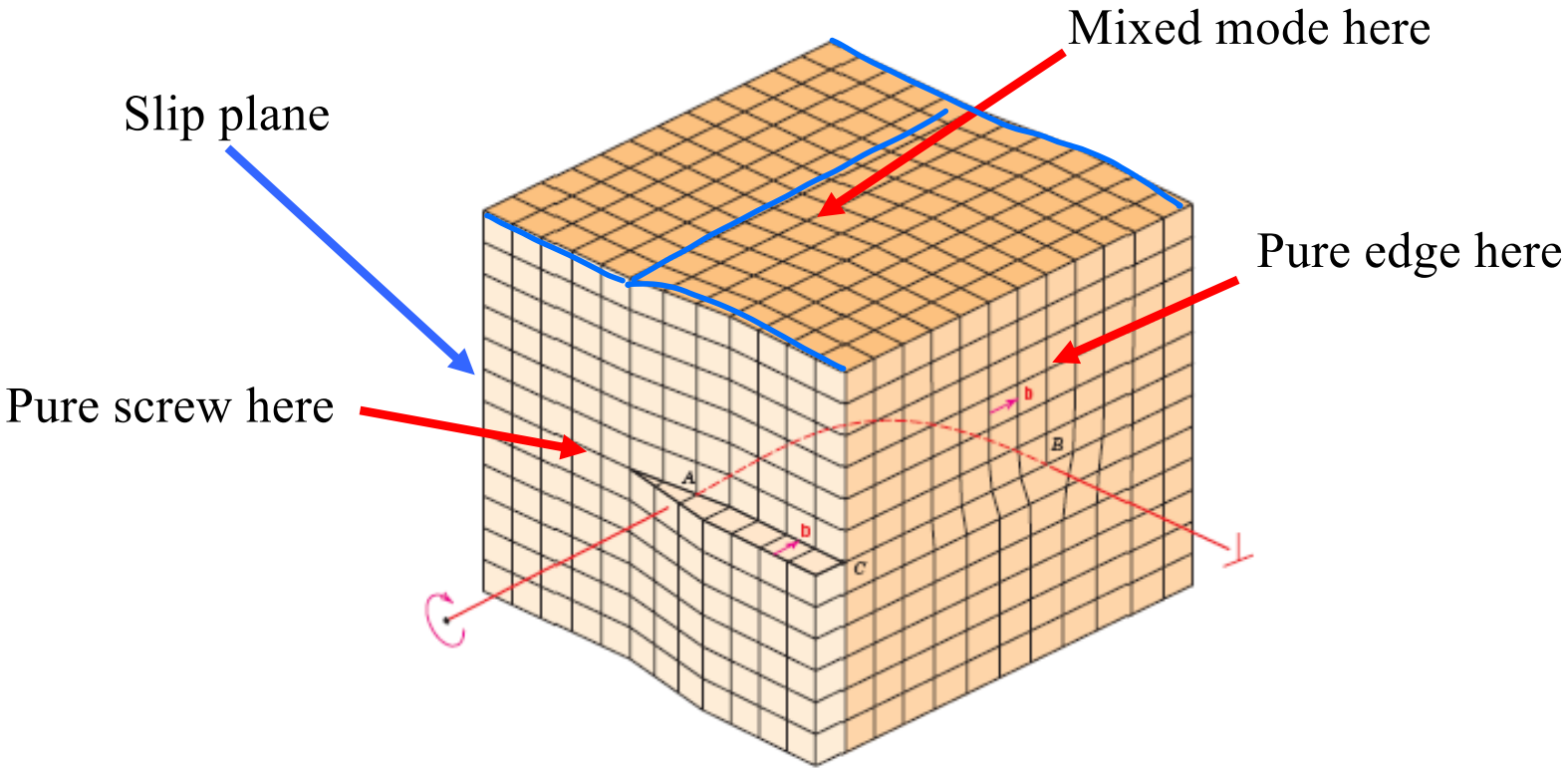




- *Crystal* is "cut halfway through and then slide sideways“
  helical path through structure hence “screw”.
- The motion of a screw dislocation can be thought of in terms of tearing a sheet of paper.

Dislocations

Usually, dislocations have both an edge and a screw character; i.e., they are ..mixed.... dislocations:



Summary

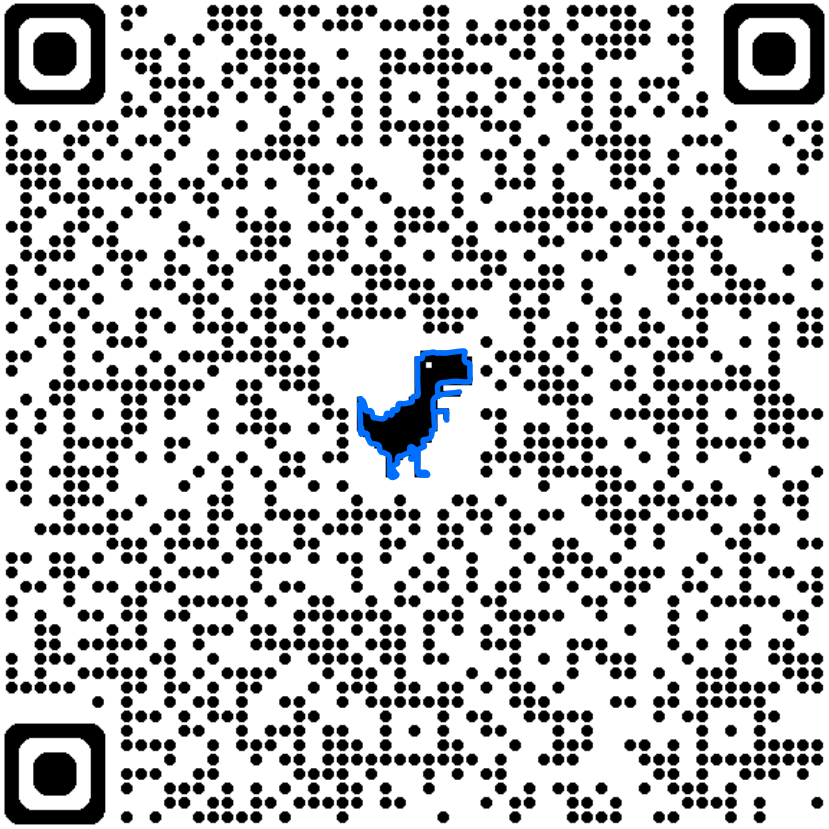
Which statement is MOST correct regarding defects in solids?

A. Defects may be either desirable or undesirable, and may be either intentionally introduced or naturally occurring.

B. All defects are undesirable because they disrupt the crystal structure.

C. Desirable defects are always intentionally introduced.

D. Unintentional defects are always harmful.



Use the QR code shown in the classroom. This one won't work.

Next time:

Grain Boundaries and Microstructure

